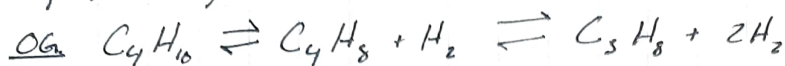


PROBLEM SET 5

121 Czerw

(6.1 c) (IMRCF)



$$K_c = 0.05 \text{ mol/dm}^3$$

$$F_{A0} = 10 \text{ mol/min}, F_{B0} = 0, F_{C0} = 0$$

$$C_{T0} = 0.2 \text{ mol/dm}^3$$

(rate of diffusion) $R_B = k_c C_B \rightarrow k_c \Delta(C_B - C_{BS}) \rightarrow C_{BS} = 0$

$$\rho_b = 1.5 \text{ g/cm}^3$$

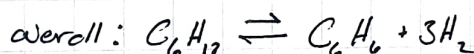
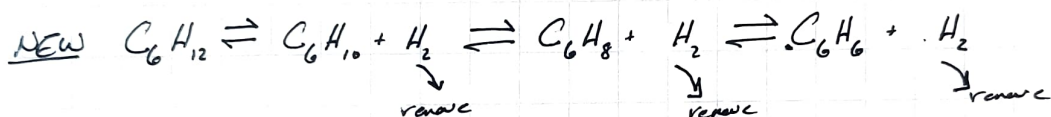
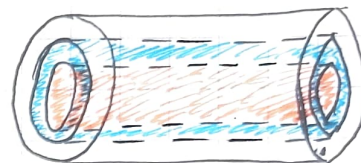
$$k = 0.7 \text{ 'min}$$

$$k_c = 0.2 \text{ 'm}$$

iso thermal & isobaric ($T_0/T = 1, p = 1$)

membrane

catalyst - 2cm Diam.



changes

$$K_c = 0.001 \text{ mol/dm}^3$$

mole balances

$$dF_A/dV = r_A, \quad dF_B/dV = r_B - R_B, \quad dF_C/dV = r_C$$

rate law

$$-r_A = k \left(C_A - \frac{C_B C_C}{K_c} \right)$$

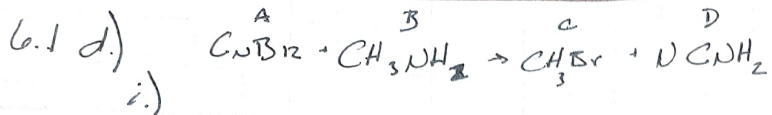
relative rates (based of each individual rxn).

$$r_A/1 = r_B/1 = r_C/1$$

Concentration

$$C_A = C_{T0} F_A/F_T, \quad C_B = C_{T0} F_B/F_T, \quad C_C = C_{T0} F_C/F_T$$

* only thing that changes in code is value of K_c *



i.)

$C_{B0} \rightarrow 2 \text{ min } (C_{B0} = 0)$ conversion, rate of rxn $\downarrow$ concentration of C_B, C_C & C_D all stay at zero over time as no reaction can happen when one of the reactants is missing. C_A does still drop though which can be accounted for by the volatility of cyanogen bromide since the concentration is measured as the concentration in solution.

$\rightarrow 2 \text{ max } (C_{B0} = 0.1 \text{ mol/dm}^3)$, 100% conversion is reached after $\approx 100 \text{ s}$. because of this C_A goes to zero @ the same time $\downarrow$ since A goes to zero the B being added has nothing to react w/ $\downarrow$ rxn rate also goes to zero. For the same reason C_B begins to rise post 100s @ the same rate it's being added. C_C & C_D also begin to drop overtime because as B is being added post depletion of A , it takes up more space in the solution causing C & D to be more spread out $\downarrow$ lowering their mol/dm^3 .

$V_0 \rightarrow 2 \text{ min } (V_0 = 0.001)$, conversion does not exceed 0.05 over the 500s as there is not enough B to react w/ all of A . For the same reason rxn rate stay very low, C_A drops only slightly, C_C & C_D rise only slightly $\downarrow$ C_B stays at zero since all of it reacts w/ A .

$\rightarrow 2 \text{ max } (V_0 = 1.000)$, has the same affect as $C_{B0} 2 \text{ max}$ w/ minor differences. instead of conversion reaching 1 right after 100s $\downarrow$ rxn rate reaching zero at same time they both reach their respective max/min @ 100s. the graph that is most different is Concentration vs. time. C_A has a sharper slope downward reaching zero right before 100s, C_C & C_D still peak $\downarrow$ then drop but peak @ $\approx 0.005 \text{ mol/dm}^3$ instead of $\approx 0.025 \text{ mol/dm}^3$ $\downarrow$ reach to almost zero by 500s instead of ≈ 0.009 but the drop happens for the same reason. C_B however instead of growing none stop does start to gradually plateau $\downarrow$ that's because the concentration of C_{B0} has a greater impact on C_B than V_0 as V_0 caps out at 1 $\downarrow$ so its maximum flow rate (mol/s) is only as high as the concentration. i.e. when $C_{B0} = 0.025$ $\downarrow$ $V_0 = 1$ then $F_{B0} = 0.025 \text{ mol/s}$ vs if $C_{B0} = 0.1 \text{ mol/dm}^3$ $\downarrow$ $V_0 = 1 \text{ dm}^3/\text{s}$ then $F_{B0} = 0.1 \text{ mol/s}$.

$V_0 \rightarrow 2 \text{ min } (V_0 = 1)$ the volume of the reactor is small, meaning that even if the concentration of A is the same (mol/dm^3) there are less moles of A in the reactor initially $\downarrow$ less B is needed to react w/ all of A , that's why the conversion reaches 100% faster (i). It's also why the rxn rate reaches its peak @ ≈ 10 seconds if $A_i = 0.05 \text{ mol/s}$ & $V_0 = 0.05 \text{ dm}^3/\text{s}$ $\downarrow$ $C_{B0} = 0.025 \text{ mol/dm}^3$, you need 0.05 moles of B to fully convert A $0.025 \times 0.05 = 0.00125 \text{ mol/s}$ you need 40s. to get enough B to react w/ all of A , $\downarrow$ then include k so get how fast it reacts.

6.1 d cont

V_0 min cont $\rightarrow$ is also why the reaction rate doesn't sky rocket $\hat{?}$ then plot it like when C_{B0} is maxed out. Instead in tops out $\hat{?}$ just under $0.0004 \text{ mol/dm}^3 \cdot \text{s}$. There isn't enough B coming in $\hat{?}$ once to react w/ A even though there is less A to react w/. The concentration graph looks similar to when V_0 is large. C_A drops to zero right around 120 s $\hat{?}$ after that C_D/C_C also starts to drop after peaking because C_B begins to take up more space.

V_0 max $\rightarrow$ ($V_0 = 40 \text{ dm}^3$) graphs follow the same trajectory as when V_0 is minimum for pretty much the same reason. There is more A in the tank $\hat{?}$ the amount of B coming into the tank is not enough to substantially react w/ A. 500 s is not enough time to get the conversion to 100% $\hat{?}$ instead only gets to 50%. C_A does deplete over time but since all of B is reacting C_B stays at zero $\hat{?}$ since B never accumulates in the tank C_C $\hat{?}$ C_D never drop as A is also being used up $\hat{?}$ the amount of total moles in the reactor does not change.

ii) a smaller V_0 (≈ 0.0052) creates the same result as a large V_0 (≈ 40).

Rxn rate constant k has the least effect on how the graphs change, final concentration of all 4 species only vary by about 0.003 mol/dm^3 from when $k = 0.2$ to $k = 5.0$ by the end of the 500 s. The molar flow rate of B has the ability to completely change all 3 relationships w/ time $\hat{?}$ as soon as there is a build up (excess) of B, concentrations of C $\hat{?}$ D will start to fall.

iii) new rate law

$$-r_A = k \left[C_A C_B - \frac{C_C C_D}{K_C} \right]$$

e) i) rxn rate constant k has largest effect when all other variables are at the amounts stated in problem, when $k = 100 \frac{\text{dm}^3}{\text{mol} \cdot \text{kg cat} \cdot \text{s}}$ $X \approx 0.04$, when $k = 15000 \frac{\text{dm}^3}{\text{mol} \cdot \text{kg cat} \cdot \text{s}}$, $X \approx 0.74$. That's a est 0.7 difference. C_{A0} $\hat{?}$ C_{B0} come close w/ a 0.6 difference between min $\hat{?}$ max values.

ii) $\alpha = 0.009$, $k = 15000$, $C_{A0} = 0.023$, $C_{B0} = 0.009$

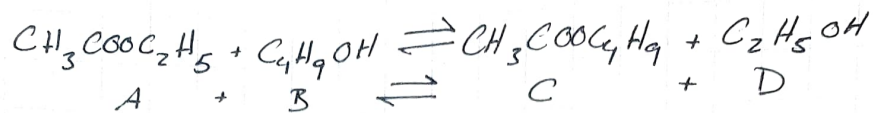
$\alpha = 0.010$, $k = 7806$, $C_{A0} = 0.010$, $C_{B0} = 0.010$

$\alpha = 0.009$, $k = 5000$, $C_{A0} = 0.025$, $C_{B0} = 0.029$

$\alpha = 0.009$, $k = 3965$, $C_{A0} = 0.050$, $C_{B0} = 0.050$

iii) the smaller the alpha (α) the less likely pressure is to drop in reactor. Unless α is zero, there will be some sort of pressure drop w/ this reaction. W/ almost any combination of variables, once p drops below 0.4, reaction rates converge to zero. Conversion varies little w/ α . Molar flow rate is proportional to concentration.

P6.11)



$$T = 300\text{K}$$

$$K_c = 1.08$$

$$k = 9 \cdot 10^{-5}$$

$$\text{Volume Ethyl Acetate} = 200\text{ dm}^3$$

$$V_{0\text{BUT}} = 0.05\text{ dm}^3/\text{s}$$

$$C_{A0} = 7.72\text{ mol/dm}^3$$

$$C_{B0} = 10.93\text{ mol/dm}^3$$

a) since K_c is large $\therefore$ k is really small, almost no conversion happens

$$\frac{\text{dm}^3}{\text{mol} \cdot \text{s}} \left(\frac{\text{mol}}{\text{dm}^3} \right)$$

b.) conversion of Ethyl Acetate (B) is very low even after 5000 s. Only about 1.3%. Rate of rxn peaks @ ~100 seconds with a rate of $0.00012\text{ mol/s} \cdot \text{dm}^3$. Concentration of Butanol (C_D) hovers just above zero, about 0.10 mol/dm^3 by the end of the 5000 s.

c.) when ethanol evaporates right away, formation of Butyl Acetate increases as the equilibrium is never reached because a product is continuously removed. Conversion also increases because of that.

d.) with how small k is, almost no amounts of F_{B0} , N_{A0} $\therefore$ V_0 will raise conversion. If k is raised $\therefore$ K_c is lowered conversion rate increases.

e.) ~~that~~ I learned how removing a product can substantially increase the conversion of reactants even if k is low $\therefore$ K_c is high. I think the point of the problem is to show how reactors can be modified to increase desired yields.

f.) would increasing the temp (or decreasing depending on if rxn is endo or exothermal) be more cost efficient (time vs money) than distilling out ethanol? Question needs to factor in cost of changing temp, how that would effect the time needed for desired reaction, would changing the temp cause any secondary side rxns to occur?